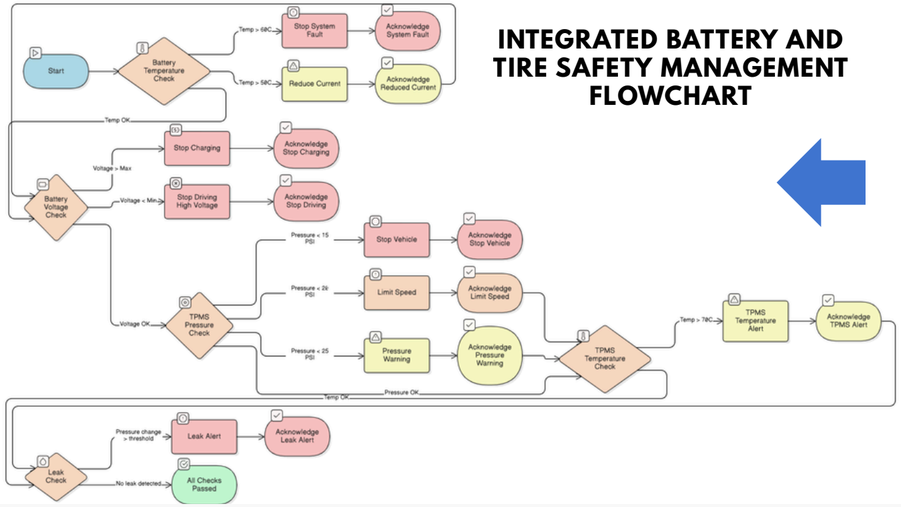
CONTROL SYSTEM UNIT

Commercial three-wheeler EVs are widely used for last-mile delivery and passenger transport, operating under heavy daily use, frequent stop-start cycles, and strict cost constraints. Key challenges include long charging times, battery degradation, safety concerns, and limited monitoring. This project aims to develop a system integrating fast charging, TPMS, and smart features to enhance efficiency, safety, and vehicle uptime.

SYSTEM ARCHITECTURE

- The proposed EV system consists of multiple interconnected subsystems:
- - Energy System: Battery pack, Battery Management System (BMS), and DC bus
- - Drive System: BLDC motor and motor controller
- - Charging System: Onboard charger and fast charging interface
- - Sensing System: TPMS, temperature, voltage, and current sensors
- - Control System: Vehicle Control Unit (VCU)
- - Communication System: CAN bus for real-time control and IoT module for cloud connectivity
- AI-Based Intelligent Monitoring and Optimization System
- The VCU acts as the central controller, managing all subsystems and enabling intelligent decision-making.



INTEGRATED BATTERY AND TIRE SAFETY MANAGEMENT FLOWCHART

FAST CHARGING SYSTEM

- The charging system supports both AC slow charging and DC fast charging. Fast charging is based on a Constant Current-Constant Voltage (CC-CV) profile. For a 6 kWh battery and 3 kW charger, charging time is reduced to approximately 2.2 hrs considering efficiency losses.
- Fast charging requires effective thermal management since heat rises with I^2R losses. Temperature sensors regulate current to prevent overheating. For this low-voltage commercial three-wheeler EV, Type-6/Type-7 are preferred for cost and compatibility, while CCS2 targets high-voltage passenger EVs.

MOTOR AND DRIVE SYSTEM

A 2–4 kW high torque geared BLDC motor is selected for its efficiency and reliability. The motor controller uses PWM-based switching to regulate speed and torque. Regenerative braking is implemented to recover kinetic energy during braking, improving overall efficiency.

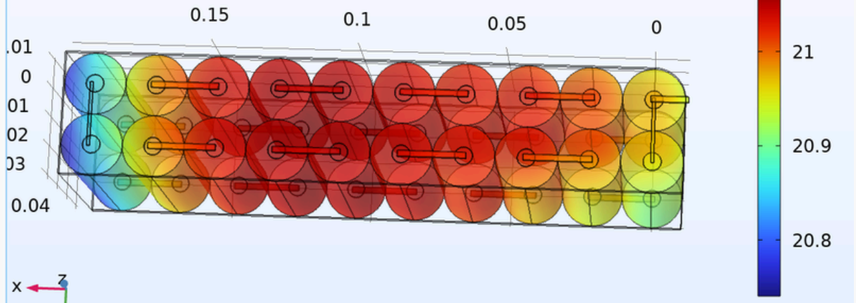
TIRE PRESSURE MONITORING SYSTEM (TPMS)

- Each wheel is equipped with a pressure and temperature sensor connected to a low-power microcontroller and wireless transmitter (BLE WiFi or RF). The data is sent to the VCU for processing.
- Key features:
- - Real-time pressure display
- - Tire position identification
- - Low-pressure warning
- - Leak detection based on rate of pressure change
- TPMS improves safety, reduces tire wear, and enhances energy efficiency

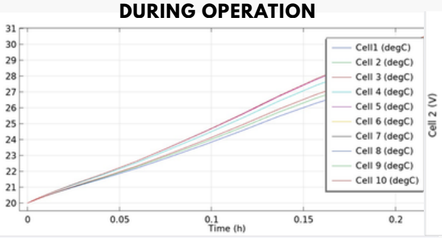
BATTERY SYSTEM DESIGN

- A lithium iron phosphate (LFP) battery is chosen for its safety, long life, and cost-effectiveness. For a 100 km range at 60 Wh/km, about 6 kWh capacity is required. Using 3.2V cells, a 16-series setup provides a 48V system, with parallel cells added for capacity.
- The BMS handles key functions such as SOC estimation (coulomb counting), cell balancing, voltage protection, and thermal management.

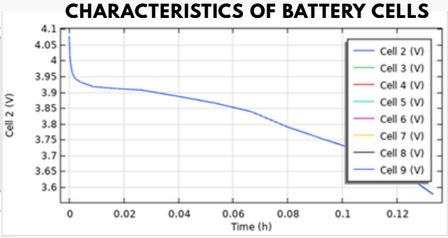
THERMAL DISTRIBUTION MODEL OF LITHIUM-ION BATTERY PACK



THERMAL PROFILE OF BATTERY CELLS DURING OPERATION



VOLTAGE DISCHARGE CHARACTERISTICS OF BATTERY CELLS



BAJAJ CAMPUS CHALLENGE 2026

THE INDUCTORS

Distributed Intelligent Control System for Electric Vehicles

BY SUDIPTA PAL
AYISHA KUMARI

REAL-TIME EV MONITORING AND CONTROL INTERFACE



VEHICLE CONTROL UNIT (VCU)

- The VCU integrates data from all subsystems and executes control logic. It operates based on a state machine with modes such as idle, driving, charging, and fault. Example logic includes:
- - Reducing charging current at high temperatures
- - Limiting speed at low SOC
- - Generating alerts for abnormal conditions

IOT AND CLOUD INTEGRATION

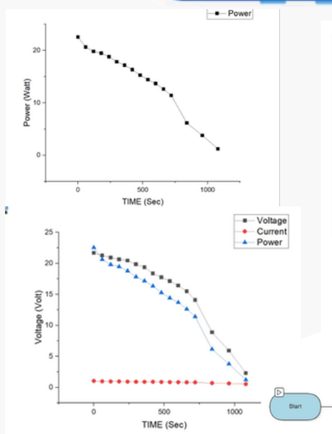
- An IoT module (ESP32 with GSM/WiFi) enables real-time data transmission to the cloud using MQTT or HTTP protocols. Data such as SOC, speed, GPS location, and tire pressure are stored and visualized using platforms like Firebase or AWS.
- This enables:
- - Fleet management
- - Predictive maintenance
- - Remote diagnostics

AI-BASED INTELLIGENT MONITORING AND OPTIMIZATION SYSTEM

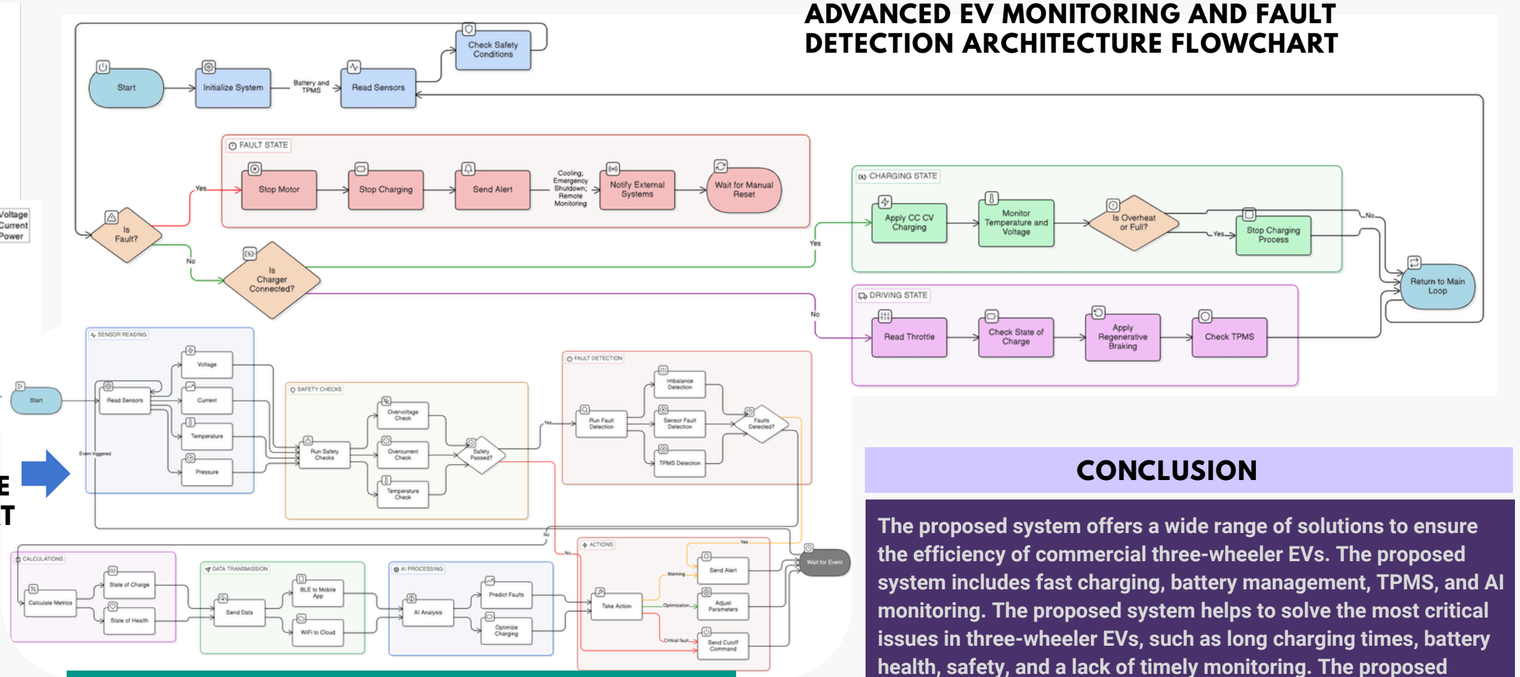
AI is integrated within the Vehicle Control Unit (VCU) and cloud platform, utilizing real-time data from the BMS, TPMS, and motor controller via CAN communication. By analyzing voltage, current, temperature, and pressure trends, it performs State of Health (SOH) estimation, predictive fault detection, and adaptive charging optimization. This enables early anomaly detection, improved range prediction, optimized energy usage, and predictive maintenance, thereby increasing vehicle uptime and operational reliability for commercial applications.

COST OPTIMIZATION STRATEGY

- The design must ensure minimum cost while maintaining performance.
- a) Hardware Cost Reduction
- Use off-the-shelf components (ESP32, standard sensors).
- Minimize the number of sensors without compromising safety.
- Use modular design so features can be added or removed based on budget.
- b) Manufacturing Optimization
- Use standard PCB designs to reduce production cost.
- Choose locally available components to reduce import cost.
- Bulk production can reduce cost by 20–30%.
- c) Software Optimization
- Use open-source platforms (Arduino, Python, Firebase free tier).
- Cloud-based AI eliminates need for expensive onboard processors.



ADVANCED EV MONITORING AND FAULT DETECTION ARCHITECTURE FLOWCHART



AI-BASED BATTERY MANAGEMENT AND VEHICLE SAFETY SYSTEM FLOWCHART

COMPONENTS	COST
LFP BATTERY PACK (6 KWH)	₹55,000-₹65,000
BATTERY MANAGEMENT SYSTEM (BMS)	₹3,000 - ₹6,000
ESP32 CONTROLLER	₹300 - ₹500
TPMS SENSORS (4 WHEELS)	₹2,000 - ₹4,000
VOLTAGE/CURRENT SENSORS	₹500 - ₹1,000
TEMPERATURE SENSORS	₹200 - ₹500
FAST CHARGING CIRCUIT	₹5,000 - ₹10,000
WIRING, PCB, ASSEMBLY	₹2,000 - ₹5,000
SOFTWARE & CLOUD SETUP	₹1,000 - ₹3,000

COST VS BENEFIT ANALYSIS

Initial Investment Increase
- Adds ~10–15% to total vehicle cost

Long-Term Savings
- Battery life increase → saves ₹20,000–₹30,000 over time
- Reduced downtime → more daily income for drivers
- Preventive maintenance → reduces repair cost

Payback period: ~6–12 months

MARKET DEMAND AND COMMERCIAL IMPACT

Growing EV Market
- India's EV market is growing at 20–30% CAGR
- 3-wheelers dominate the EV segment (last-mile delivery, e-rickshaws)

Demand for Smart Features
Fleet operators and drivers demand:
- Real-time vehicle monitoring
- Predictive maintenance
- Reduced operating cost
- Increased uptime

Problems in Current Market
- Battery failures
- Long charging times
- Lack of monitoring
- Safety issues

CONCLUSION

The proposed system offers a wide range of solutions to ensure the efficiency of commercial three-wheeler EVs. The proposed system includes fast charging, battery management, TPMS, and AI monitoring. The proposed system helps to solve the most critical issues in three-wheeler EVs, such as long charging times, battery health, safety, and a lack of timely monitoring. The proposed system optimizes energy efficiency, predictive maintenance, and operational efficiency.

The proposed system requires a slightly higher investment, but the advantages in terms of battery longevity, low maintenance, and reliability ensure a quick payback. The rise of the EV market with its increasing demand for smart and connected vehicles makes the proposed system highly suitable. The proposed system is highly efficient in commercial applications.

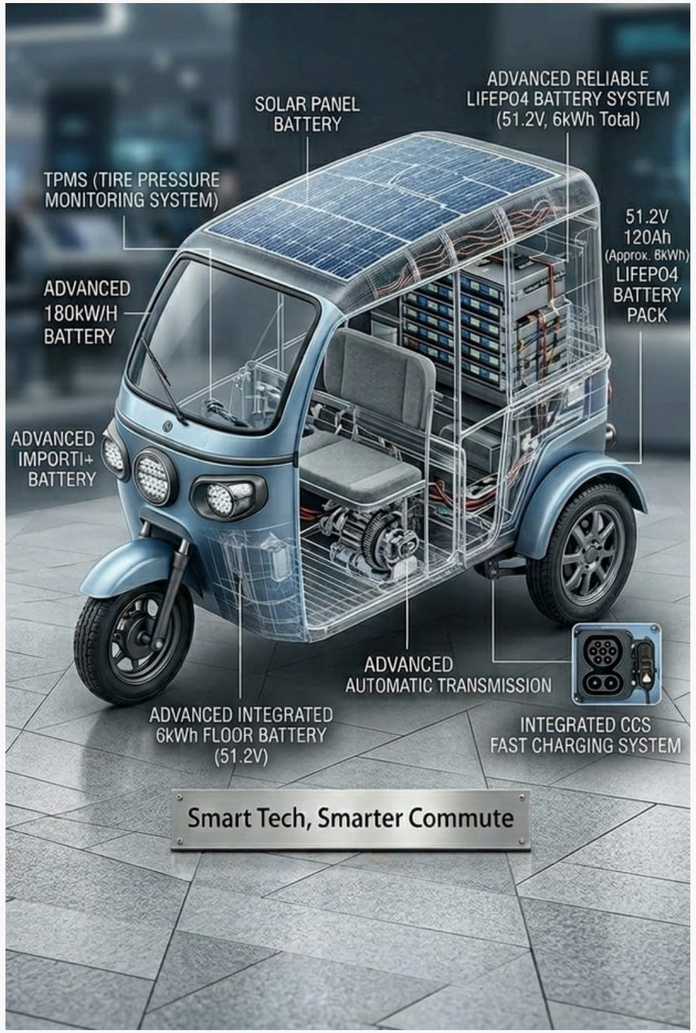
THE CORE GAP

- The biggest issue is the lack of an integrated solution.
- Current EV systems work in isolation (charging, battery, safety, monitoring are separate).
- There is no unified system that simultaneously handles:
- Charging optimization
- Battery health management
- Tire safety
- Remote diagnostics
- Additionally, solutions must be low-cost and scalable, especially for commercial vehicles.
- This creates a gap where advanced features exist, but not in a single affordable system.

INTRODUCTION

Commercial three-wheeler EVs operate under high daily utilization and strict cost constraints. Key challenges include long charging time, battery degradation, limited real-time monitoring, and safety risks.This proposal presents a Distributed Intelligent Control System (DICS) integrating fast charging, TPMS, intelligent control, and cloud connectivity to improve uptime, safety, and operational efficiency.

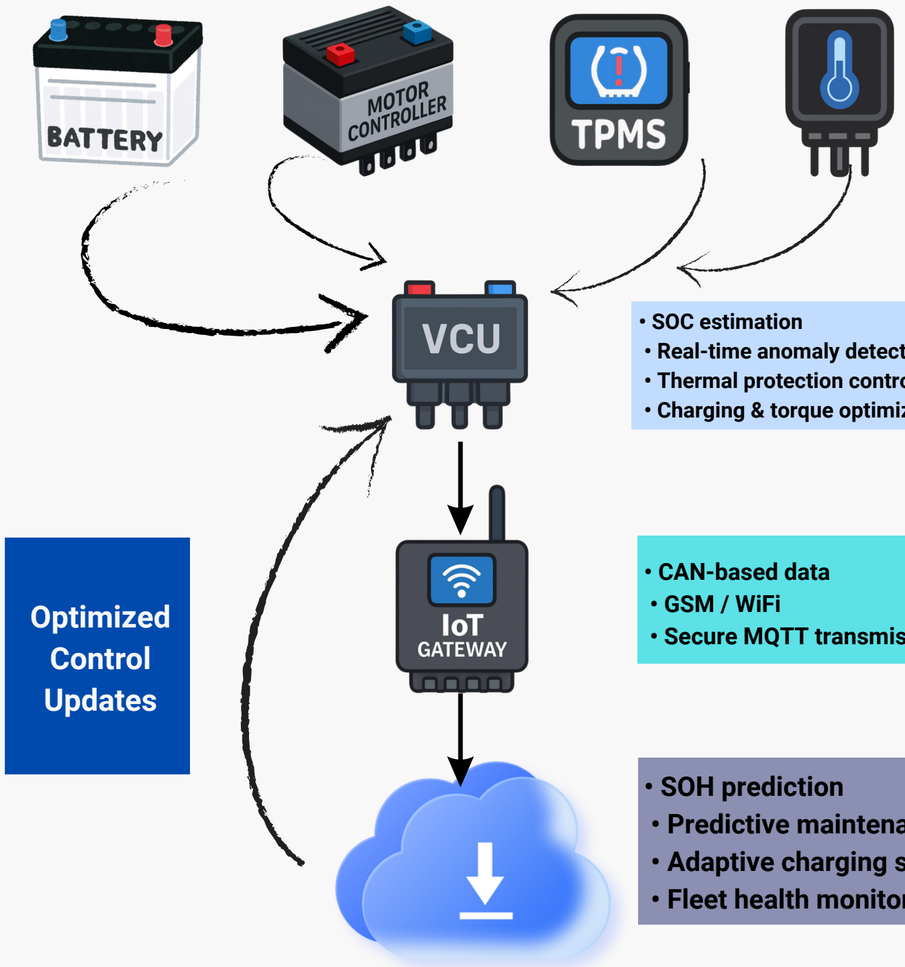
COMPONENTS	SPECIFICATIONS	COST(IN ₹)	FEATURES
LFP Battery Pack	48V, ~120Ah, 6 kWh	55,000 – 65,000	High safety, long life (3000+ cycles), thermal stability
Battery Management System (BMS)	16S, balancing, protection	3,000 – 6,000	Overvoltage, undervoltage, overcurrent protection, SOC estimation
TPMS Sensors	4 sensors, RF 433 MHz	2,000 – 4,000	Real-time tire pressure & temperature monitoring
Drive System	2–4 kW BLDC 48V, 50–100A motor controller	30,000 – 40,000	High torque efficiency PWM control, regenerative braking
Fast Charging Circuit	3 kW CC–CV charger	5,000 – 10,000	Reduced downtime Current derating, temp-based protection
Software & Cloud Setup	Firebase / AWS IoT	1,000 – 3,000	Centralized optimization Rule-based + AI-assisted control,Fleet-level analytics Remote diagnostics, predictive alerts



KEY FEATURES

- Intelligent VCU Control – Centralized real-time control of all subsystems
- AI-Based Monitoring – Fault prediction, SOH estimation, range prediction
- IoT & Cloud Connectivity – Real-time data, fleet monitoring, remote diagnostics
- Fast Charging – CC–CV charging with reduced charging time
- Advanced BMS – SOC estimation, protection, and cell balancing
- Safety System – Overvoltage, overcurrent, temperature protection
- TPMS – Real-time tire pressure and temperature monitoring
- Energy Optimization – AI-based efficient power and charging control
- Predictive Maintenance – Early fault detection and reduced downtime
- Thermal Management – Prevents overheating and improves battery life
- Cost-Effective & Scalable – Suitable for commercial EV applications

Intelligent Monitoring & Connected EV Architecture



- SOC estimation
- Real-time anomaly detection
- Thermal protection control
- Charging & torque optimization

- CAN-based data
- GSM / WiFi
- Secure MQTT transmission

- SOH prediction
- Predictive maintenance
- Adaptive charging strategy
- Fleet health monitoring

Optimized Control Updates

FUNCTIONAL SAFETY LAYER

- Over-voltage / under-voltage cutoff
- Over-current protection
- Thermal shutdown threshold
- Short-circuit isolation

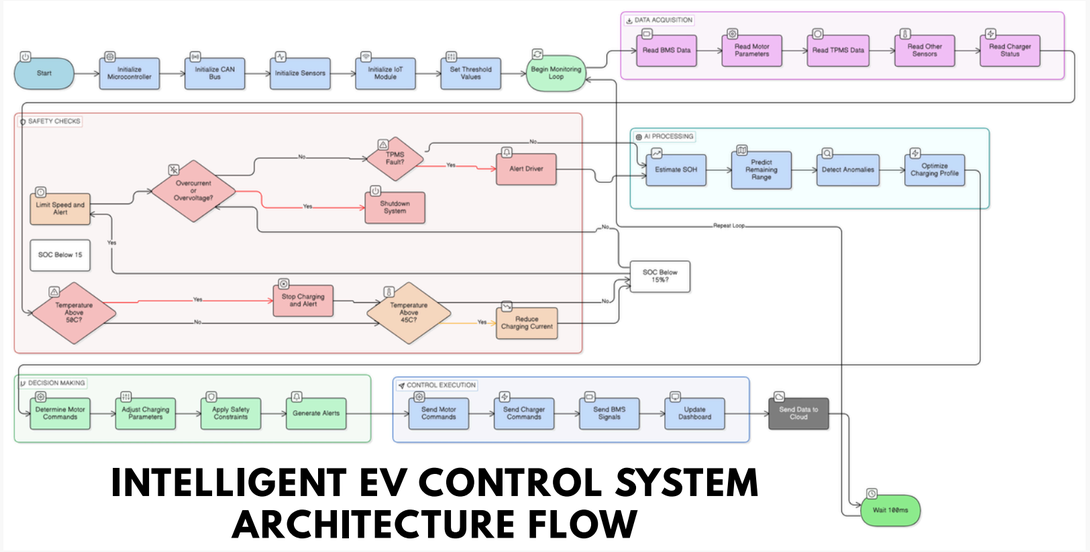
Protection logic remains active even if AI/cloud fails.

Overall Working Flow

Sensors → VCU → IoT Module → Cloud → AI Analysis → Decision → Action

REAL-TIME EV MONITORING AND CONTROL INTERFACE



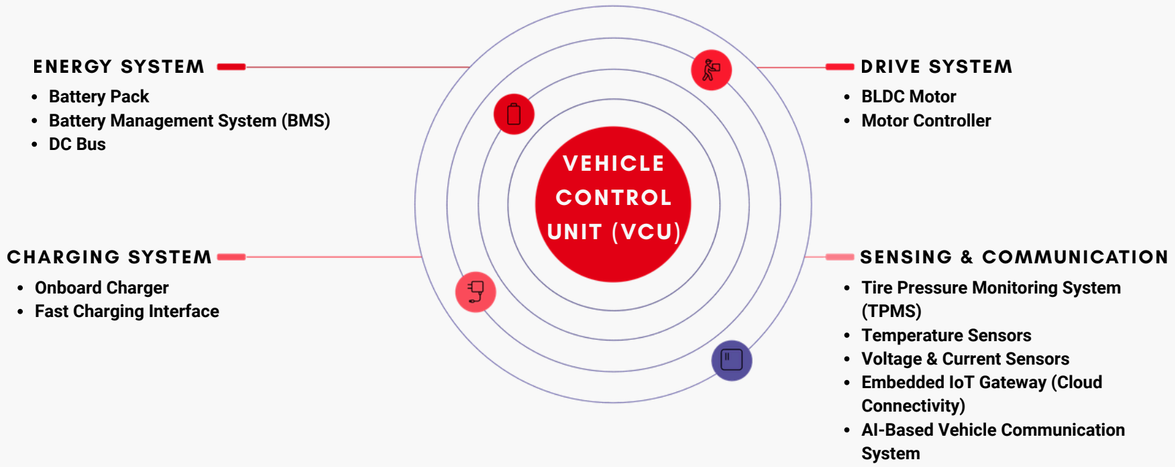


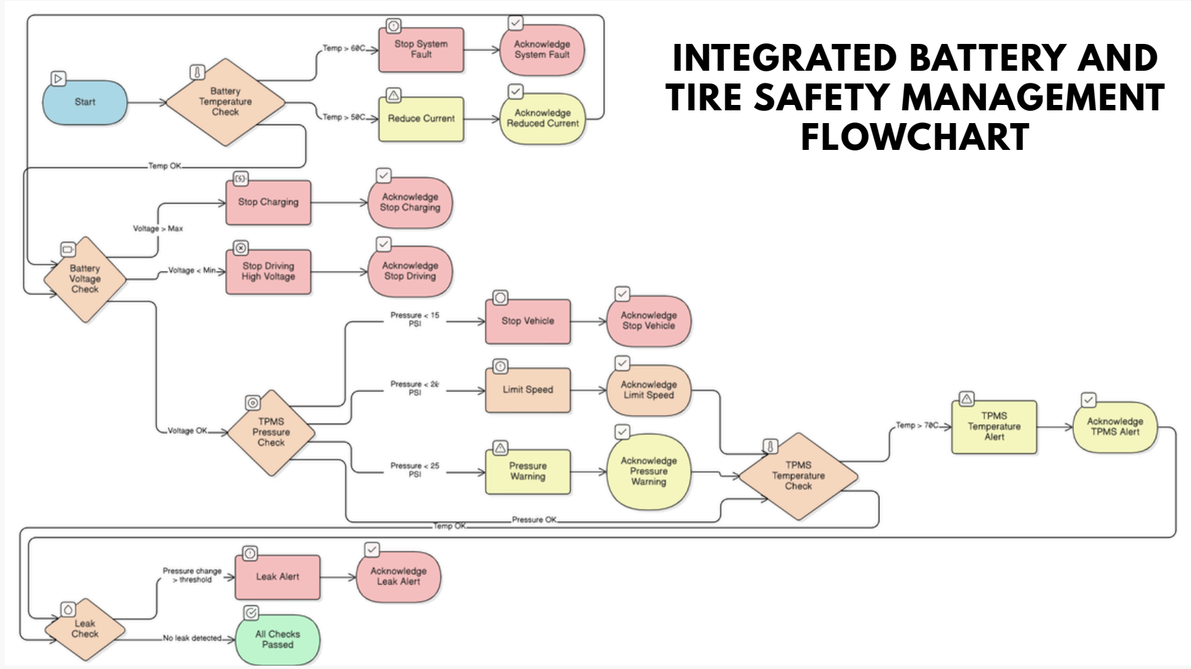
INTELLIGENT EV CONTROL SYSTEM ARCHITECTURE FLOW

SYSTEM IMPACT

- Increased uptime
- Extended battery life
- Reduced unexpected failures
- Real-time fleet visibility
- Scalable architecture suitable for fleet-level deployment.

SYSTEM ARCHITECTURE AND DRIVE SYSTEM



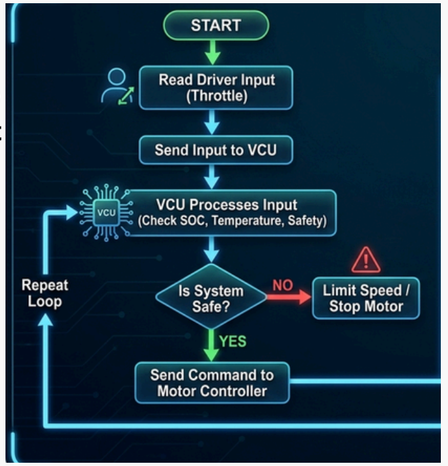


DRIVE SYSTEM

The drive system is responsible for converting electrical energy from the battery into mechanical motion to move the vehicle. It mainly consists of a BLDC motor and motor controller. The motor controller receives commands from the VCU and regulates the motor speed and torque based on driver input and operating conditions. The BLDC motor provides high efficiency, smooth operation, and low maintenance. The system also supports features like speed control, torque control, and regenerative braking, improving overall performance and energy efficiency.

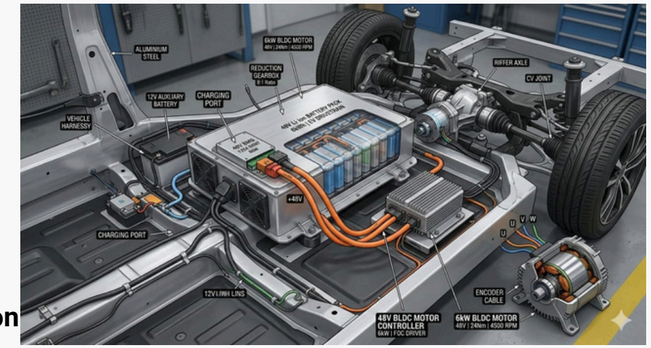
Working Principle

1. Battery supplies DC power to the motor controller
2. Motor controller converts DC to controlled AC (PWM signals)
3. Controller adjusts current and voltage based on throttle input
4. BLDC motor rotates and generates mechanical torque
5. Power is transmitted to wheels through the drive mechanism
6. VCU monitors system and controls speed, torque, and safety



Drive System: Features & Advantages

- BLDC Motor – High efficiency, low maintenance, smooth operation
- Motor Controller – Controls speed/torque, enables regenerative braking
- DC-DC Converter – Powers 12V electronics from battery
- Solar + MPPT – Supports auxiliary charging, improves efficiency
- Cooling System – Prevents overheating, increases battery life
- VCU – Central control for intelligent and safe operation
- GPS + Display – Real-time tracking and user information
- CAN Module – Fast and reliable communication
- Relay/Contactor – Ensures electrical safety



VEHICLE CONTROL UNIT (VCU): THE DECISION-MAKING CORE

DATA ACQUISITION

- The VCU collects critical real-time data from all subsystems via the CAN bus, creating a complete picture of vehicle status:
- Battery Management System (BMS): SOC, SOH, voltage, current, temperature
- Motor Controller: speed, torque, power
- Sensors: temperature, current, voltage, IMU data
- TPMS: tire pressure and temperature
- Charging System: charging status and current

DATA PROCESSING

- The collected data is processed using a combination of rule-based and AI-based algorithms:

Rule-Based Control

- Ensures fast and reliable safety actions
- Thermal protection (over-temperature control)
- Overcurrent / overvoltage shutdown
- Low SOC speed limiting

AI-Based Control

- State of Health (SOH) estimation
- Range prediction
- Anomaly / fault detection
- Adaptive charging optimization

REAL-TIME DECISION MAKING

- Based on processed data, the VCU makes instantaneous decisions:
- Safety: stop charging at high temperature, trigger alerts
- Performance: control motor torque and speed
- Energy Optimization: manage power and charging efficiently

All decisions are executed within milliseconds

CONTROL EXECUTION VIA CAN BUS

1.Motor Controller → Speed & Torque Control

- VCU controls motor speed and torque based on driver input and battery condition
- Limits speed at low SOC or high temperature

2. Charger → Charging Control

- Adjusts charging current according to battery temperature and health
- Stops charging in unsafe conditions

3. BMS → Battery Management

- Monitors voltage, current, and temperature
- Activates protection during faults

4.Dashboard → Alerts & Display

- Shows SOC, range, and system status
- Gives warnings like low battery or faults

CLOUD CONNECTIVITY (IOT INTEGRATION)

1.Communication Protocols

- Uses lightweight protocols like MQTT (real-time, low bandwidth) and HTTP (web communication)
- Ensures reliable data transfer even on low network conditions

2.Data Transmission

- Sends real-time vehicle data such as:
- SOC & SOH (battery status)
- Voltage, current, temperature
- Faults and alerts
- TPMS data (tire condition)
- Vehicle location (GPS)
- Provides a complete live status of the vehicle

3.Cloud Platforms

- Data is stored and processed on platforms like Firebase or AWS IoT
- Enables scalable and secure data management

SMART FEATURES ENABLED

1. Fleet Monitoring & Tracking
   - Allows operators to monitor multiple vehicles in real time from a central dashboard
   - Displays parameters like SOC, location, speed, battery status, and faults
   - Helps in route planning and improving operational efficiency
   - Enables full visibility and control of the entire fleet
2. Predictive Maintenance
   - Uses historical and real-time data to predict possible failures
   - Detects early signs of issues like battery degradation, overheating, or tire wear
   - Alerts are generated before a breakdown occurs
   - Reduces unexpected downtime and maintenance cost
3. Remote Diagnostics
   - Vehicle faults and performance data can be analyzed remotely without physical inspection
   - Engineers can identify issues such as sensor failure, battery faults, or system errors
   - Reduces the need for frequent manual checks
   - Saves time and operational effort

PSEUDO CODE

```
Initialize system

while(vehicle ON)
{
    // Data Acquisition
    read_BMS();
    read_motor();
    read_TPMS();
    read_sensors();

    // Safety Checks
    if(temp > 50) stop_charging();
    else if(temp > 45) reduce_current();

    if(SOC < 15) limit_speed();
    if(overcurrent) shutdown_system();

    // AI Processing
    SOH = estimate_SOH();
    range = predict_range();
    detect_faults();

    // Decision Making
    control_motor();
    control_charging();

    // Communication
    send_CAN_commands();
    send_data_to_cloud();

    delay(100ms);
}
```

TIRE PRESSURE MONITORING SYSTEM

The Tire Pressure Monitoring System (TPMS) continuously measures the pressure and temperature of each tire and displays the values in real time, as shown in the given image where all tires are within the normal range (around 30–32 psi). For a typical three-wheeler EV, the front tire usually operates in the range of 28–32 psi, while the rear tires, which carry more load, operate in the range of 32–36 psi. If the pressure drops below the normal range, the system categorizes it into warning and critical levels to ensure safety.

- Normal Pressure: Front (28–32 psi), Rear (32–36 psi) → safe operation
- Warning Pressure: Front < 26 psi, Rear < 30 psi → efficiency reduces, alert required
- Critical Pressure: Front < 22 psi, Rear < 26 psi → high risk, immediate action needed
- Normal Temperature: 30°C – 50°C → safe range
- Warning Temperature: > 60°C → indicates overheating
- Critical Temperature: > 70°C → risk of tire burst
- Importance: Maintaining proper pressure improves safety, increases battery efficiency, reduces tire wear, and prevents sudden failures



TIRE	PRESSURE	STATUS
FRONT LEFT	32 PSI	NORMAL
FRONT RIGHT	30 PSI	NORMAL
REAR LEFT	31 PSI	SLIGHTLY LOW
REAR RIGHT	32 PSI	NORMAL

PSEUDO CODE

```
void initialize_TPMS(){
while(vehicle_ON)
{
    for(int j=0;j<tyre;j++) //for each tyre
    {
        P = read_pressure();
        T = read_temperature();
        rate = (P - P_prev)/time;

        // Pressure Check
        if(P < critical_pressure)
            alert("Critical Pressure");
        else if(P < warning_pressure)
            alert("Low Pressure");

        // Temperature Check
        if(T > 70)
            alert("High Temp - Critical");
        else if(T > 60)
            alert("High Temp");

        // Leak Detection
        if(rate < -5)
            alert("Fast Leak");
        else if(rate < -2)
            alert("Slow Leak");

        send_to_VCU(P, T, rate);
        P_prev = P;
    }
    delay(2000);
}
}
```

FAST CHARGING SYSTEM

System initialization & battery connection check → Continuous monitoring of voltage, current & temperature → Charging stops immediately under abnormal conditions

- **Input Voltage**:230V AC Single Phase Grid Compatible
- **CC Mode:** High current charging (voltage rises gradually)
- **CV Mode:** Voltage held constant, current tapers down
- Charging terminates when current drops below set threshold
- Real-time data transmitted to VCU & cloud for monitoring and optimization

Average Charging time :
2.2 Hours to Full Charge

BATTERY DESIGN & CALCULATION (48V LFP SYSTEM)

1. Energy Requirement Calculation

- Required range = 100 km
- Energy consumption = 60 Wh/km

Total Energy Required=100×60=6000 Wh=6 kWh

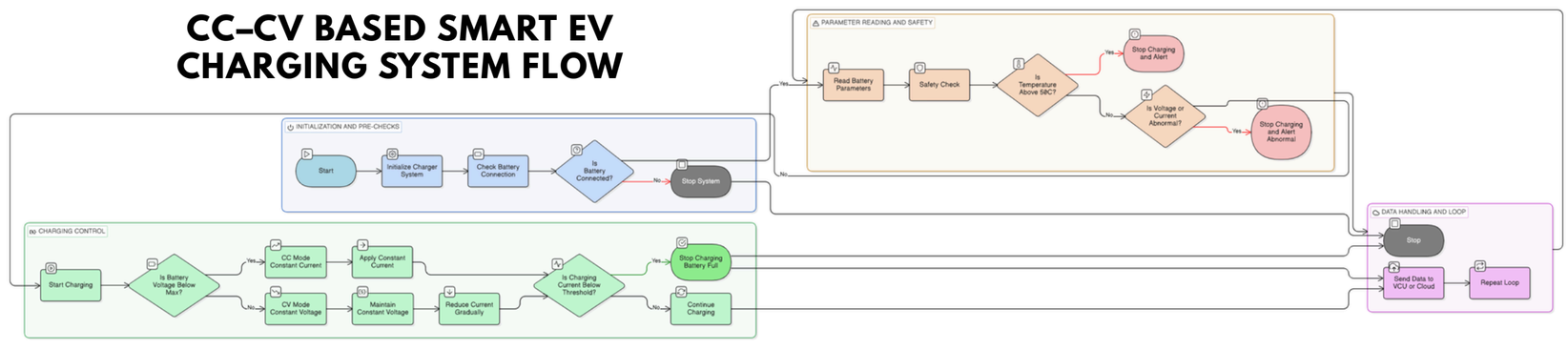
2. Battery Voltage Calculation

- Single LFP cell voltage = 3.2 V
- Number of cells in series = 16

Pack Voltage=3.2×16=51.2
Nominal system ≈ 48 V battery

3.Capacity (Ah) Calculation

Capacity (Ah)=Energy (Wh)/Voltage (V)=6000/51.2≈117 Ah
Required battery capacity ≈ 120 Ah



ALGORITHM

Fast Charging System (CC–CV Method)

1. Initialization
   - Charger, sensors, and control unit are activated
   - System checks if battery is connected
   - If not connected → charging stops
2. Data Monitoring & Safety Check
   - Reads battery parameters:
     - Voltage
     - Current
     - Temperature
   - If temperature > 50°C → stop charging
   - If voltage/current abnormal → stop charging
3. Constant Current (CC) Mode — Fast Charging
   - Charger supplies fixed high current
   - Battery voltage increases gradually
   - Provides fast charging (≈70–80%)
4. Transition to CV Mode
   - When battery reaches maximum voltage
   - Charger switches from CC to CV
5. Constant Voltage (CV) Mode — Safe Charging
   - Voltage is kept constant
   - Current gradually decreases
   - Prevents overcharging and overheating
6. Charging Completion
   - When current falls below threshold
   - Battery is fully charged
   - Charging is stopped
7. Continuous Monitoring
   - Parameters are monitored throughout the process
   - Safety checks are always active
8. Communication with VCU & Cloud
   - Data sent to VCU and cloud
   - Enables monitoring and control

THERMAL ANALYSIS

Introduction to COMSOL

COMSOL Multiphysics is a finite element analysis (FEA) software used to simulate real-world physical systems. It can model multiple physics domains such as heat transfer, electrical, fluid, and chemical systems together, making it suitable for complex applications like battery packs.

Why COMSOL is Used?

COMSOL enables accurate simulation without physical experiments, saving time and cost. It provides 3D visualization and supports both steady-state and time-dependent analysis, helping in better design and optimization.

Features Relevant to the Project

COMSOL includes modules like Heat Transfer for temperature analysis and Electrochemistry for battery behavior. Its multiphysics capability allows coupling of thermal and electrical effects, while post-processing tools provide clear visualization of results.

Relevance to Battery Bank

Battery packs generate heat during operation, and the heat distribution is often non-uniform. COMSOL helps simulate temperature profiles, identify hot spots, and design cooling systems to maintain safe operating conditions (20–40°C).

Heat Management in Batteries

Heat is generated due to internal resistance and reactions, given by Q=I2R Excess heat reduces efficiency, shortens battery life, and can cause safety issues. Proper thermal management is essential for safe operation.

Battery Pack Structure

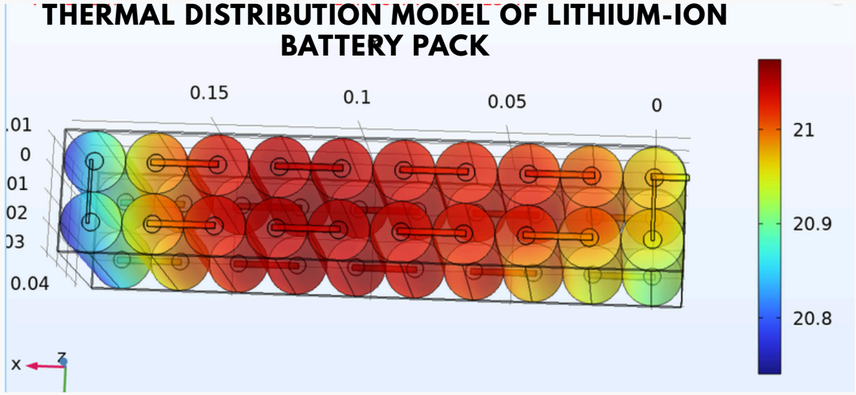
Battery packs consist of cylindrical cells arranged in a structured layout. Inner cells tend to be hotter, while outer cells remain cooler due to better cooling. This non-uniformity must be considered in design.

Electrochemical Heating

Heat in batteries comes from reversible (entropy-based) and irreversible (loss-based) processes. Both contribute to temperature rise and affect battery performance and efficiency.

Heat Flux Boundaries

Heat flux boundaries define how heat leaves the battery. Convective heat transfer is given by q=h(T−T∞). Proper boundary conditions are important to simulate realistic cooling and thermal behavior.



This demonstrates the thermal analysis of a battery pack using COMSOL, focusing on how heat is transferred from the battery to its surroundings. Batteries generate heat during operation due to internal resistance, and understanding this heat flow is essential for safe and efficient EV performance.

Convective Heat Flux

Convective heat flux represents the heat loss from the battery surface to the surrounding air. In COMSOL, this is applied as a boundary condition where heat is transferred based on the temperature difference between the battery surface and the ambient air.

In the image, the color distribution shows that:

- Inner cells are hotter (yellow/red)
- Outer cells are cooler (blue/green)

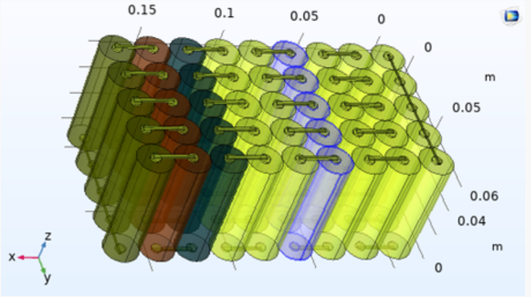
This happens because outer cells are exposed to air and can dissipate heat more easily. This model simulates real-world cooling, such as air cooling in EVs.

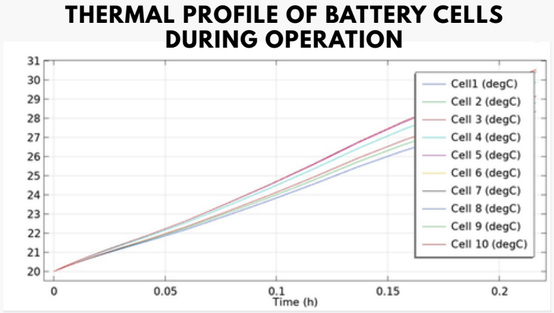
Thermal Insulation

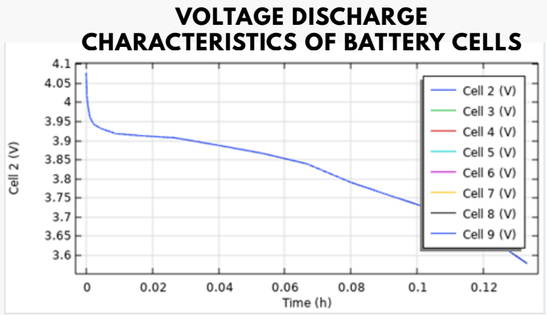
Thermal insulation represents a condition where no heat is allowed to escape from the battery pack. In COMSOL, this is modeled by setting the heat flux to zero at the boundaries.

- Heat accumulates inside the battery pack
- Inner cells become much hotter

This simulates situations where batteries are enclosed without proper cooling, leading to heat buildup and potential overheating.







Importance in EV Systems

Battery packs in EVs generate heat during charging and discharging. If this heat is not removed:

- Battery life decreases
- Efficiency reduces
- Risk of thermal runaway increases

Therefore, cooling systems (air or liquid cooling) are required to maintain temperature within a safe range (around 25–40°C).

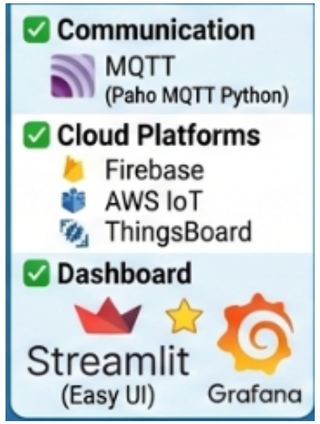
AI-BASED INTELLIGENT MONITORING+ IoT + CLOUD INTEGRATION IN EV SYSTEM

The integration of IoT (Internet of Things) and Artificial Intelligence (AI) in electric vehicles enables the development of smart, connected, and autonomous systems. IoT provides real-time data collection and communication, while AI analyzes this data to make intelligent decisions. Together, they transform EVs from simple electric machines into intelligent, data-driven mobility systems.

Role of IoT in EV Systems

IoT enables real-time connectivity and data exchange between the vehicle, cloud, and user.

- Collect data from sensors (battery, motor, TPMS, GPS)
- Transmit data to cloud using protocols like MQTT/HTTP
- Enable remote monitoring and control
- Support fleet management systems



Role of AI in EV Systems

AI processes real-time and historical data to provide intelligent insights and decision-making.

- Predict battery health (SOH estimation)
- Detect faults and anomalies
- Predict vehicle range
- Optimize charging and energy usage
- Enable adaptive control

Integration of IoT and AI

Working Flow:

- Sensors collect real-time data
- IoT module sends data to cloud
- Cloud stores and processes data
- AI analyzes data
- Decisions are made
- Commands sent back to vehicle

AI / Machine Learning Tools & Working

Programming & Development Environment

- Language: Python
- Tools:
- Google Colab / Jupyter Notebook → model training
- VS Code → full system development

Used for data processing, ML, and AI implementation

Data Transmission (IoT Layer)

- ESP32 + MQTT protocol is used
- Send data to cloud (Firebase / AWS IoT)

Real-time communication between vehicle and cloud



Data Collection (Hardware Layer)

- Sensors collect real-time data:
- Voltage
- Current
- Temperature
- Tire Pressure

Data is read using ESP32 / Microcontroller (C/C++)

Machine Learning Models

Use:

- Scikit-learn
  - Regression → SOH prediction
  - Classification → Fault detection
- TensorFlow / PyTorch
  - Time-series prediction
  - Deep learning models

Predict battery health and faults

Data Processing (Python)

- Use Python libraries:
- NumPy → numerical calculations
- Pandas → data
- handling
- Matplotlib / Seaborn → visualization

Deployment (Real System)

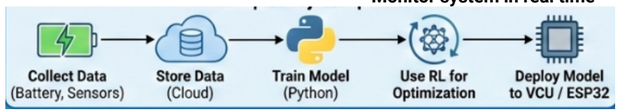
- Convert model using:
  - TensorFlow Lite / ONNX
- Deploy on:
  - ESP32 / VCU

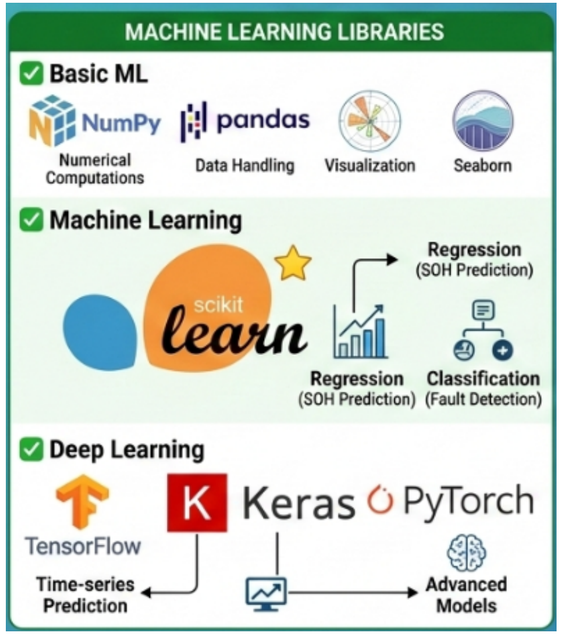
Enables real-time AI control

Visualization & Monitoring

- Use:
  - Streamlit → simple dashboard
  - Grafana → advanced visualization

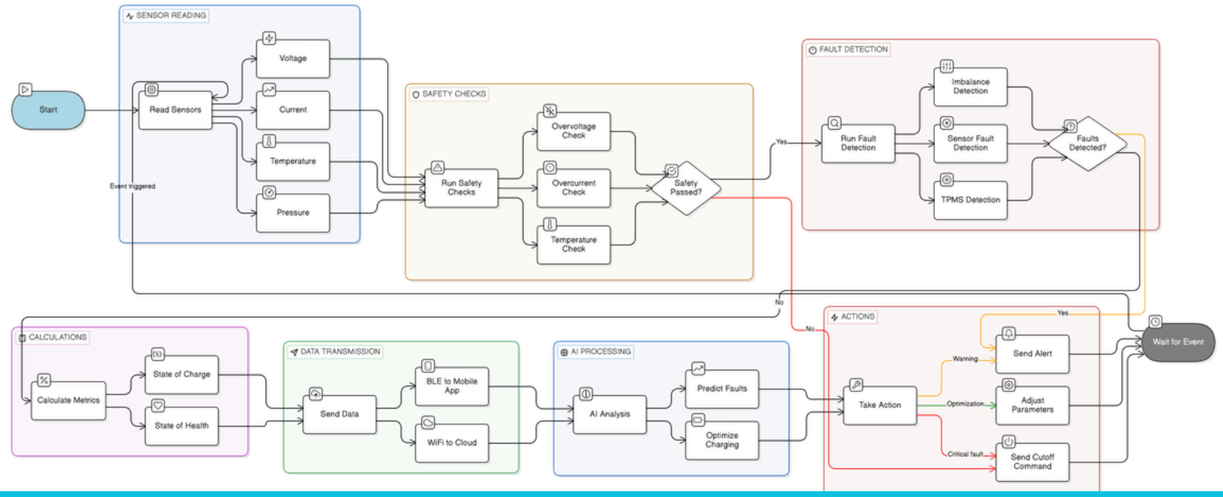
Monitor system in real time





Advantages of AI + IoT Integration

- Improved Safety
- Early fault detection
- Real-time alerts
- Optimized charging
- Better energy management
- Reduced Maintenance Cost
- Predictive maintenance
- Avoids unexpected failures
- Extended Battery Life
- Controlled charging
- Temperature management



OPPORTUNITY OF INDIA'S EV LANDSCAPE

Accelerating EV Adoption & Policy Push

•India is one of the fastest-growing EV markets
• Strong adoption in 2W, 3W & commercial fleets
• Government push via FAME & state EV policies

Economic & Operational Shift Toward EVs

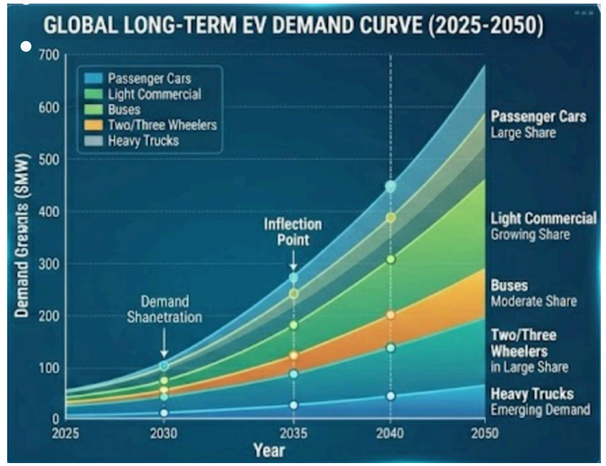
•Rising petrol & diesel prices accelerating transition
• Lower operating & maintenance costs vs ICE vehicles
• Higher uptime directly improves fleet revenue

Demand for Intelligent & Sustainable Mobility

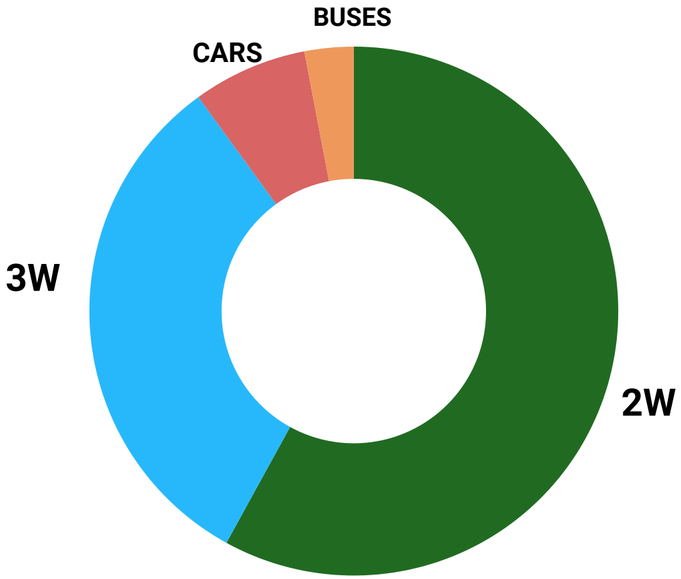
• Need for real-time monitoring & predictive maintenance
• EVs reduce emissions and support clean mobility goals

KEY CHALLENGES IN CURRENT EV ECOSYSTEM

- Long Charging Time
- Battery Degradation
- Safety Risks
- Lack of Real-Time Monitoring



INDICATIVE DISTRIBUTION BASED ON RECENT EV ADOPTION TRENDS IN INDIA



3W COMMERCIAL VEHICLES HAVE HIGHER DAILY UTILIZATION AND STRONGER NEED FOR UPTIME OPTIMIZATION, MAKING THEM IDEAL FOR INTELLIGENT MONITORING SYSTEMS

Green Technology & Renewable Support by EV

1. Reduction of Carbon Emissions
2. Energy Efficiency
   - EVs have higher efficiency (~80–90%) compared to ICE vehicles (~30–40%)
3. Integration with Renewable Energy
4. Reduced Dependence on Fossil Fuels
5. Sustainable Transportation System

CONCLUSION

This project presents an AI-based intelligent monitoring and control system for electric vehicles, integrating IoT, cloud computing, and advanced control strategies. The system enables real-time data acquisition, safety monitoring, and intelligent decision-making using machine learning and predictive algorithms.
By incorporating features such as battery management, fast charging, TPMS, and thermal monitoring, the system enhances vehicle safety, improves energy efficiency, and extends battery life. The integration of IoT and cloud platforms allows remote monitoring, predictive maintenance, and fleet management, making the system suitable for commercial EV applications.

India's commercial EV segment represents a high-growth opportunity for intelligent, fast-charging low-voltage systems.

---------- THANK YOU ----------

THANK YOU

THE PROBLEM: COMMERCIAL EVS ARE HOLDING BACK THE LAST MILE

⚡ LONG CHARGING TIME

- Commercial three-wheeler EVs typically use AC slow charging, which takes around 6–8 hours for a full charge.
- These vehicles operate in high-utilization environments (delivery, passenger transport), where maximum uptime is required.
- Long charging cycles lead to idle vehicles, reducing the number of trips per day.
- This directly impacts fleet revenue and driver earnings.

🔋 BATTERY DEGRADATION

- The battery is the most expensive component of an EV, and its health determines vehicle performance.
- Frequent deep discharge, overcharging, and high temperature operation accelerate battery aging.
- Poor thermal management increases internal resistance, leading to heat generation and faster degradation.
- Over time, battery capacity reduces, resulting in lower range and efficiency.
- Battery replacement is costly, so degradation significantly increases the Total Cost of Ownership (TCO).

⚠️ SAFETY RISKS

- Safety is a critical concern in commercial EVs, especially under heavy usage.
- Lack of fault detection systems can result in unnoticed electrical or thermal failures.
- Battery overheating may lead to fire hazards or system shutdown.
- Improper tire pressure can cause reduced stability, higher energy consumption, and accidents.
- Without continuous monitoring, small issues can turn into major safety incidents.

📡 NO REAL-TIME MONITORING

- Most existing systems do not provide live data tracking to operators.
- Fleet managers cannot monitor battery status, vehicle health, or location in real time.
- Maintenance is usually reactive (after failure), rather than predictive.
- This leads to unexpected breakdowns, longer repair times, and operational inefficiency.
- Real-time visibility is essential for smart fleet management and reliability.

THE CORE GAP

- The biggest issue is the lack of an integrated solution.
- Current EV systems work in isolation (charging, battery, safety, monitoring are separate).
- There is no unified system that simultaneously handles:
- Charging optimization
- Battery health management
- Tire safety
- Remote diagnostics
- Additionally, solutions must be low-cost and scalable, especially for commercial vehicles.
- This creates a gap where advanced features exist, but not in a single affordable system.

AI-BASED INTELLIGENT MONITORING+ IoT + Cloud Integration in EV System



PROGRAMMING LANGUAGE (Python ⭐)

Easy for ML, IoT, & Data Analysis

DEVELOPMENT TOOLS

- For Training Modeis & Testing
- Google Colab & Testing
- For Training Models & Testing
- For Full Project Development

MACHINE LEARNING LIBRARIES

Basic ML

- NumPy — Numerical Computations
- pandas — Data Handling
- Visualization
- Seaborn

Machine Learning

scikit learn ⭐

- Regression (SOH Prediction)
- Classification (Fault Detection)

Deep Learning

- TensorFlow
- Keras
- PyTorch

Time-series Prediction

Advanced Models

REINFORCEMENT LEARNING LIBRARIES (Important)

Adaptive Charging Control | Energy Optimization | Driving Strategy

RL

⭐ **Stable-Baselines3 (Best for You)** ⭐

- Easy to use
- Pre-built Algorithms

PPO | DQN | A2C

Best for EV Control Optimization

Gymnasium

State= SOC/Temp | Charging Current

Environment for RL Training

RLlib (Advanced)

TensorFlow Agents / PyTorch RL

AI MODELS YOU CAN USE

Battery Health Prediction

Linear Regression | Random Forest | LSTM (Time-series)

Fault Detection

Isolation Forest | SVM | Autoencoder

Reinforcement Learning (Key Part)

Problem: Optimize Charging Current → State (Advanced) → RL Setup (Action, Reward) → Agent learns optimal strategy

IOT + CLOUD TOOLS

Communication

- MQTT (Paho MQTT Python)

Cloud Platforms

- Firebase
- AWS IoT
- ThingsBoard

Dashboard

- Streamlit (Easy UI)
- Grafana

SIMULATION & DEPLOYMENT TOOLS

Simulation Tools

- MATLAB / Simulink
- Python Simulation (Custom) — Needed for RL training

For Embedded System

- ESP32 (MicroPython / C++)

Example Project Pipeline:

Collect Data (Battery, Sensors) → Store Data (Cloud) → Train Model (Python) → Use RL for Optimization → Deploy Model to VCU / ESP32

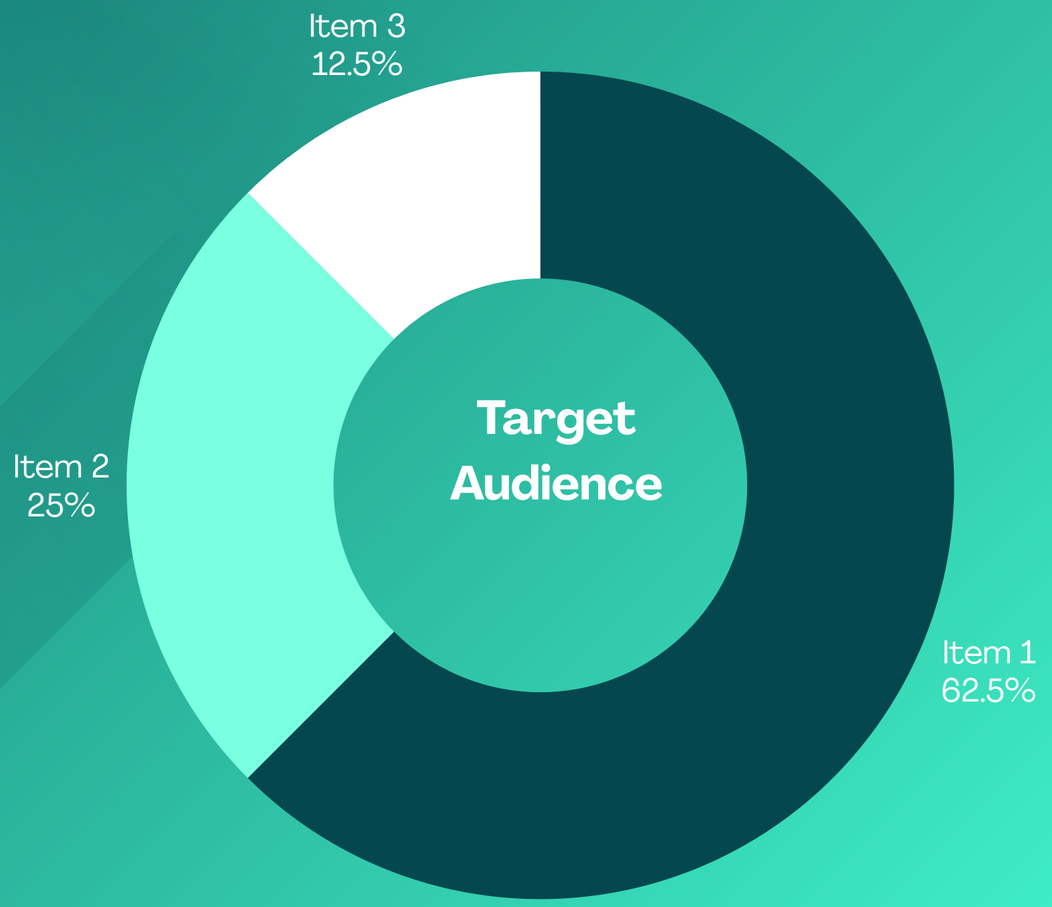
⭐ ADVANTAGES OF AI IN EV CARS ⭐

- **Enhanced Battery Longevity** through precise SOH prediction and optimized charging control [cite: 2, 6, 8, 10].
- **Optimized Range** via advanced energy management and intelligent driving strategy [cite: 2, 6, 8, 10].
- **Proactive Fault Detection** enabling predictive maintenance and higher safety [cite: 2, 6, 8, 10].
- **Improved Driving Experience** with personalized adaptive control [cite: 2, 6, 8, 10].
- **Efficient Charging** optimizing grid interaction [cite: 2, 6, 8, 10].

Market Analysis

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🔹 1. Data Acquisition (Sensors)

The system collects real-time data from sensors such as voltage, current, temperature, tire pressure, and vehicle location. This data represents the actual condition of the battery and vehicle and forms the base for monitoring and decision-making.

🔹 2. Local Processing (VCU + Safety Checks)

The Vehicle Control Unit (VCU) processes the data and performs rule-based safety checks such as overvoltage, overcurrent, and temperature limits. If any parameter exceeds safe limits, immediate actions like alerts or shutdown are triggered.

🔹 3. IoT Communication

The processed data is transmitted to the cloud using an IoT module (ESP32/TCU) through protocols like MQTT or HTTP. This enables real-time communication between the vehicle and remote systems.

🔹 4. Cloud Storage & Monitoring

The cloud stores data from multiple vehicles and provides real-time dashboards for monitoring parameters like SOC, SOH, temperature, faults, and location. This enables centralized fleet management.

🔹 5. AI-Based Data Analysis

AI analyzes both real-time and historical data to identify patterns and system behavior. It performs tasks such as battery health estimation, range prediction, fault detection, and charging optimization.

🔹 6. Predictive Fault Detection

AI predicts potential failures such as overheating, voltage imbalance, or tire issues by analyzing trends. This allows preventive action before actual faults occur, improving safety and reliability.

🔹 7. Optimization of Performance

AI optimizes charging, power usage, and energy management based on battery condition and usage patterns. This improves efficiency, extends battery life, and enhances vehicle performance.

🔹 8. Intelligent Decision Making

Based on AI analysis, the system decides the best action:

- Adjust parameters (optimization)
- Send alerts (warnings)
- Cut off system (critical fault)

🔹 9. Alerts & Remote Monitoring

Real-time alerts are sent to the driver or cloud platform for issues like low battery, high temperature, or faults. Fleet operators can monitor vehicle status remotely.

🔹 10. Predictive Maintenance

AI uses historical data to identify early signs of component failure, reducing unexpected breakdowns and maintenance costs.

🔹 11. Adaptive Control

The system dynamically adjusts motor speed, charging current, and power output based on real-time conditions, ensuring optimal performance and safety.

🔹 12. Cutoff & Protection Mechanism

In critical situations like overheating or severe faults, the system automatically shuts down to prevent damage and ensure safety.

India's Electric Vehicle (EV) market is one of the fastest-growing in the world, driven by government policies, rising fuel costs, urbanization, and environmental concerns. This creates a significant opportunity for innovative solutions, especially in the commercial vehicle segment such as three-wheelers.

🔹 1. Rapid Market Growth

India's EV market is expected to grow at a high CAGR (~25–30%), driven by increasing demand for affordable and sustainable transport. The commercial EV segment, particularly three-wheelers, is leading this growth due to its widespread use in last-mile delivery and passenger transport.

👉 This creates a large market for smart EV technologies and control systems

🔹 2. Dominance of Three-Wheeler EVs

Three-wheelers account for a major share of EV adoption in India because:

- Low cost compared to cars
- High usage in urban and semi-urban areas
- Suitable for last-mile connectivity

👉 Makes it an ideal segment for implementing intelligent EV systems like yours

🔹 3. Government Support & Policies

The Indian government is strongly promoting EV adoption through:

- FAME-II scheme (subsidies)
- EV30@30 target (30% electrification by 2030)
- State-level incentives and tax benefits

👉 Encourages investment and innovation in EV technology

🔹 4. Need for Smart Fleet Solutions

With the growth of delivery services and ride-hailing:

- Fleet operators require real-time monitoring
- Need for predictive maintenance
- Demand for higher uptime and efficiency

👉 Your AI + IoT solution directly addresses this need

🔹 5. Rising Fuel Costs

Increasing petrol and diesel prices are pushing users towards EVs due to:

- Lower running cost
- Higher efficiency

👉 EV adoption becomes economically attractive

🔹 6. Environmental Benefits

- Transportation contributes significantly to CO_2 emissions
- EVs produce zero tailpipe emissions

👉 Supports India's goal of clean and sustainable mobility

🔹 7. Challenges Creating Opportunities

India's EV ecosystem still faces issues:

- Long charging time
- Battery degradation
- Lack of monitoring systems
- Safety concerns

👉 These challenges create opportunities for:

- Fast charging systems
- AI-based battery management
- IoT-based monitoring systems

🔹 8. Role of AI and IoT in Future EVs

Future EVs will require:

- Intelligent energy management
- Predictive maintenance
- Fleet analytics

👉 AI + IoT will be key technologies driving next-generation EV systems